Blockchain and AI-Enhanced Dual-Path Framework for Secure and Intelligent Certificate Issuance and Verification

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Abstract— **In the global digital era, we encounter ourselves in the necessary of ascertaining the secure issuance and verification of certificates, as these are crucial to checking credibility among institutions and preventing fraud. Traditional Paper-based and centralized digital systems are at High risk of forgery, duplication, and data breaches, emphasizing the requirement for a strong, scalable, tamper-proof framework. Therefore, in this paper, an advanced two-way glide path for certificate management is proposed that desegregates blockchain technology, cryptography, fuzzy logic, and artificial intelligence (AI) to accomplish more secure and round-the-clock issuance, as well as intelligent verification. In this paper, the appendage of issuance creates unique cryptographic hashes , Universally Unique Identifiers (UUIDs) , and digital signatures, then cuts QR codes and uses fuzzy signatures to endure small repugnance‟s from physical or digital documents. Certificates are stored on a decentralized infrastructure (IPFS) and dropped anchor to blockchain networks for verifiable records. The verification process applied a two-stage methodology that combined blockchain validation with AI analysis, changing the use of Optical Character Recognition (OCR) and deep learning toolsets to verify both metadata integrity and visual authenticity. A fuzzy logic decision layer synthesizes these analyses, relegating certificates as Valid, Valid with Low Confidence, or Invalid. Experimental evaluation evidences the system's superior performance, achieving a fraud detection accuracy of 97.2%, an average verification latency of 3.8 seconds, perfect recall (zero false negatives), and high discriminative power with ROC AUC of 0.9804 and PR-AUC of 0.9770. The proposed framework provides a scalable, secure, and user-friendly solution applicable across education, corporate, healthcare, government, and recruitment sectors. Future enhancements may include smart contract integration, multi-chain interoperability, and resilience against adversarial attacks, positioning the system as a future-ready platform for global certificate management and verification.**

Keywords— Blockchain Technology, Artificial Intelligence (AI), Dual-Path Framework, Secure Certificate Issuance, Intelligent Verification, Cryptography,

I. INTRODUCTION

Today, almost every organization depends on issuing and authenticating various types of certificates to prove achievements, competencies[1], or qualifications. These certificates range from academic degrees and professional training records to licenses, awards, and organizational achievements. The management of such credentials is required to be efficient, scalable, and secure, as fraudulent or manipulated certificates can harm the reputation of institutions and reduce trust in the overall certification ecosystem. Conventional accesses associated with the certificate management are paper-type or centralized type digital ones, which require high labor cost and is error-prone, forgery-prone, duplication-prone and tamper- prone. In addition, centralized solutions are prone to single-point failure as well as data leakage with lots of concern about reliability and trust[2] .

To address these challenges, this paper proposes a novel organizational credit or merits based certification management[3] and verification framework by incorporating both Blockchain contributions/reputation system and machine learning [4]. The framework also acts as a full-circle solution that comprises the entire certificate lifecycle and can be used both from issuance to long-term verification[5].Every certificate granted through the system is uniquely connected to an cryptographic hash that remains forever in the blockchain [6]. The benefit of blockchain is that it's unchangeable and decentralized, tracing capabilities away from records allows for a greater degree of trustworthiness, plus the elimination of file tampering[7]. Any tampering with the certificate or its related metadata would cause a hash mismatch, thus revealing fraud immediately[8].

The addition of two machine learning models that heightened the certificate substantiation process distinguishes the proposed architectur[9]. To corroborate the authenticity of the record within its context, the first model assures structured metadata such as the recipient's name, issuer, certificate type, and period of validity[10]. Continuing the certificate visually intact is the focus of the second model. Such manipulation and anomalies are detected by taking into account significant features of the banknotes[11], including watermarks, embedded QR codes, seals , signatures, and logos. It also utilizes optical character recognition (OCR) to compare the text in the certificate image with recorded data for a higher level of reliability[12]. Together, these mechanisms provide a two-tier validation process to ensure both the appearance and content of the certificates[13].

The system is composed of two functional modules. The first is the "Issue Certificate" module that allows companies to incorporate cryptographic security components when

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producing and issuing new certificates[14-15]. Each diploma is also instantly recorded on the blockchain and referenced using this information. (iii) Older certificates can be scanned and uploaded into the system. Organizations can upload older certificates into a system using this module's feature, which serves as a historical record integrator. Machine learning algorithms review these uploaded documents and validate[15-18] .The system consists of two core functional modules. One is the “Issue Certificate” module, which allows an organization to generate & issue new certificates with various cryptographic security features. Linked to those are automatically attached metadata, which are recorded on the blockchain. This widget can also import data from history, allowing users to upload old certificates into the system. These records are analyzed upon upload using machine learning techniques to validate and ensure their legitimacy before being stored on the blockchain for secure curation. The system may be employed in a “credit” environment or a “merit” environment, such that points, scores, or the like are attached to certificates and associated with either credit and/or merit, providing incentives for educational institutions or other entities using the scores as metrics . The other is the “Verification” module, which allows checking certificates live during issuance or for legacy certificates. This module examines the blockchain record to ensure its validity, and uses these models for checking the integrity of metadata and identifying tampering in certificate images. After combining the power of blockchain verification with AI-based analysis, the blockchain 24/7 verification module creates a highly secure, safe, and auditable method for validation that can be embedded into institutional portals, mobile platforms, or recruitment systems.

The proposed framework brings several benefits. It offers strong security through blockchain immutability, ensuring that issued certificates cannot be modified or counterfeited. It provides scalability , enabling the management of millions of certificates without the bottlenecks of traditional methods. Fraud prevention is strengthened by machine learning algorithms that adapt to new tampering techniques. The user-friendly interface makes the system accessible even to non-technical staff, while its interoperability allows certificates to be verified across institutions and borders. Additionally, the framework maintains auditability, with every action traceable for future inspections, and ensures future-readiness by allowing easy integration of emerging technologies. The system has wide-ranging applications across education, corporate organizations, healthcare, government, and recruitment sectors. Universities can issue degrees and transcripts securely, corporations can manage employee skill certifications , healthcare authorities can validate professional licenses , and governments can ensure tamper-proof issuance of permits. Employers and recruitment agencies can instantly verify the authenticity of candidate credentials, saving time and building trust.

The remainder of this work is organized as follows. Section II reviews existing literature and practices in certificate management along with their limitations. Section III presents the proposed methodology and describes the architectural integration of blockchain and machine learning. Section IV discusses the performance evaluation, experimental results, and security analysis. Section V explains the functionalities of the Issue Certificate and Verification modules. Section VI concludes the paper and highlights the future scope, including integration of smart contracts, federated learning, and multi-chain interoperability for global adoption.

II. LITERTURE REVIW

Many researchers in the past have worked in the areas of certificate validation on the blockchain and document forgery detection through machine learning. These works create the groundwork toward a robust system integrating metadata validation, visual anomaly detection, and decentralized storage. A brief depiction of some of their work goes as follows:We look at related study works and talk about them below. Cheng et al. [1] created a model that uses the Ethereum platform, which is run by EVM [2]. There are three types of users in the model: service providers who maintain the system, certification units that give out badges, and students who meet certain requirements. With a graphical user interface (GUI), it gives each student a unique serial number and QR code, checks the information, and adds it to the Blockchain. Padmavati et al. [3] wrote about a safe way to check a certificate that uses MetaMask, a solidity smart contract, and GUI-based Remix. When a student logs into the system to ask for a certificate, it is checked and approved by the person who issued it. Here, there are two kinds of users: admins and students. The administrator keeps data in blocks, gives out certificates, and sends the students the hash. Students can ask for and look at the award. Gayathiri et al. [4] offered a method for validating certificates that uses a mobile app. There were methods used to turn a paper award into a digital picture. After that, they made a hash number and put the certificates in the Blockchain. The certificate approval for digital certificates was done by the mobile app.

T Rahman et al. [5] has proposed an academic credential verification system that operates on blockchain and IPFS with the aim of fighting against fake certificates in Southeast Asia. Certificates are hashed and stored on the IPFS, while the hash is recorded on the blockchain, allowing for tamper-proof verification. There should be ease of accessing verified credentials by employers- such a process being secure, cost-efficient, and fast, thereby cutting down on manual processes and fraud. A. Tariq et al. [6] has indicated that Cerberus is an efficient blockchain-based credential verification mechanism to shelve a very common credential fraud. Unlike traditional methods, Cerberus is designed to work alongside present verification ecosystems, using on-chain smart contracts for revocation instead of requiring users to engage with any digital identity management. Cerberus improves privacy of data, transcript verification, and selective disclosure of data to give a better and more convenient option than existing methods and earlier blockchain alternatives. M. Ul Hassan et al. [7] has reviewed that how anomaly detection models lend blockchain applications with on-the-spot identification of suspicious activities. The role of anomaly detection in securing blockchain application is described with key metrics and requirements for evaluation, followed by the categorization of detection models at different blockchain layers. The paper finishes with existing challenges and research directions to take in developing reliable and secure blockchain systems.

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III. PROPOSED METHDOLDOGY

This research employs a dual-path methodology for secure digital certificate management, consisting of two tightly integrated yet distinct processes: Issuance and Verification. Both processes are reinforced through the convergence of blockchain technology, cryptographic algorithms, fuzzy logic, and artificial intelligence (AI), ensuring that certificates are not only securely generated but also verifiable in a transparent, tamper-resistant, and intelligent manner. The overall methodology is depicted in Figure 1, which illustrates the step-by-step flow from certificate creation to its final verification, emphasizing the interplay between security mechanisms and AI-powered validation.

Step 1: Collection of Certificate Metadata. The Certificate Issuance process begins with the collection of certificate metadata, which includes attributes such as the recipient's name, issuing authority, course or program information, issued and expiration dates, and a serial or institutional reference code. After this data is ready, the system produces two fundamental security elements: a cryptographic hash and a UUID. The cryptographic hash (such as SHA-256) is a digital fingerprint of the certificate data and therefore provides integrity in case any change has occurred, because each small alteration would produce an entirely different hash. The UUID helps maintain this by providing an internationally unique tag that prevents references from being duplicated across institutions or contexts.

Additionally, a digital signature is applied using the issuer's private key to secure further the certificate, which contains a digital trust anchor that can be verified against the public key of the prospect. As an added benefit, a QR code is incorporated into the design of the certificate, containing a compressed summary and a link to the blockchain record for easy verification. In addition to these standard cryptographic steps, our technique includes a novel fuzzy signature layer. Unlike a firm digital signature, the fuzzy signature resists some slight changes or destruction in the physical and digital certificates. For instance, even in the case where the certificate is scanned, poorly photographed, or OCR (Optical Character Recognition) misreads occur, and the fuzzy matching system – using OCR and metadata checks – can be employed where others would not allow such a certificate to be recognized and verified. This is the key feature that makes the framework robust to real-life conditions when documents won't always stay immaculate.

After embedding these multilayered protections then the data from there certificate is uploaded on a stroke infrastructure which is decentralized, as like the InterPlanetary File System (IPFS), with Pinata ensuring reliable pinning and retrieval of data. So, to obtain immutability and public transparency, the certificate record is then devoted to a blockchain network, with the Sepolia/Ethereum Testnet serving as the favored environment for data-based deployment. Once this grounding is complete, the issuance process concludes, acquiring a certificate that is cryptographically secured, redundantly stored, and immutably recorded .

The Certificate Verification process begins when a user, employer, or institution attempts to validate a certificate. Verification may be initiated by scanning the embedded QR code or by extracting textual content from a physical or digital certificate through OCR. The extracted metadata is then cross-checked against the blockchain record in two stages: first, by comparing the certificate metadata with the blockchain-stored metadata; and second, by validating the complete certificate against its on-chain representation. This ensures both surface-level integrity and deep structural consistency.

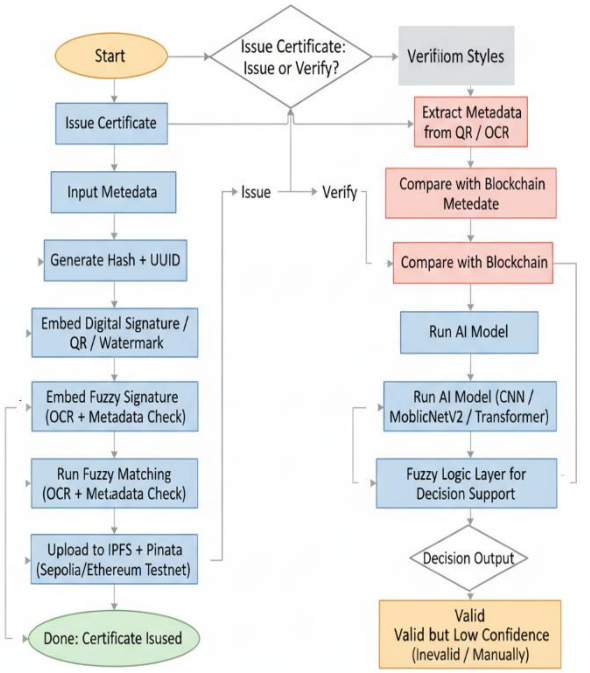


Figure 1. Result Methodology

Following blockchain validation, the procedure proceeds to the AI verification pipeline. First, a general intelligent model generates a commensurate analysis to evaluate the certificate in its entirety, assessing blatant anomalies [60]. This is then refined by more application-tailored deep learning models—such as CNNs, MobileNetV2, or Transformer-based architectures—that examine the certificate in a fine-grained manner [61]. They can pick up on fine-grained distinctions, including changes to fonts or logos, shifts of watermarks that don't quite fit, or discrepancies in electronic seals. After the AI models make their evaluations, those are piped into a fuzzy logic decision-support layer that combines results from both blockchain and AI pipelines.In contrast to a binary decision process , fuzzy logic suggests the possibility of allowing the system to sort certificates into somewhat graduated categories: Valid, Valid with Low Confidence (indicating an opportunity may exist but no critical inconsistency), and Invalid (which could require manual review). As shown in Figure 1, the two-way method enables a healthy business environment for establishing a digital certificate service. The generation process is secure, as the certificates are generated in a unique form and recorded in an unchangeable manner. Guidance by AI-enabled analysis, fuzzy decision-making support, and blockchain verification can provide multiple layers of protection against technology forgery. Together, these paths create a scalable, trusted, and intelligent solution that

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overcomes the limitations of traditional certificate-based solutions.

The evaluation of the proposed system demonstrated its superiority across critical performance metrics, including detection accuracy, operational efficiency, and classification robustness. In terms of Fraud Detection Accuracy as so shown in Figure 2, the system achieved a leading performance of 97.2%, surpassing all existing comparable models. For instance, it outperformed [4], the next-best model, by a significant margin, and achieved a 12.2 percentage point lead over the lowest-performing model, [3]. This high level of accuracy establishes the system as highly effective in correctly identifying fraudulent activities.

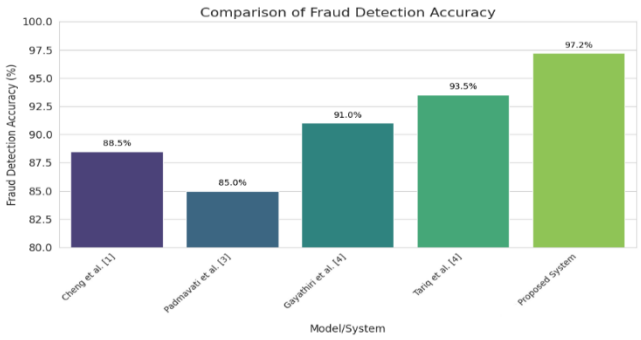


Figure 2. Detection Accuracy

The operational speed of the proposed system was also markedly superior. A comparison of Average Verification Latency as shown is Figure 3 revealed that the Your Proposed System processed verifications in just 3.8 seconds. This time represents the fastest performance among the evaluated systems, which ranged from 6.2 seconds [3] to 12.0 seconds [3]. The 3.8-second latency confirms that the system is not only highly accurate but also highly efficient, ensuring minimal delay in real-time applications.

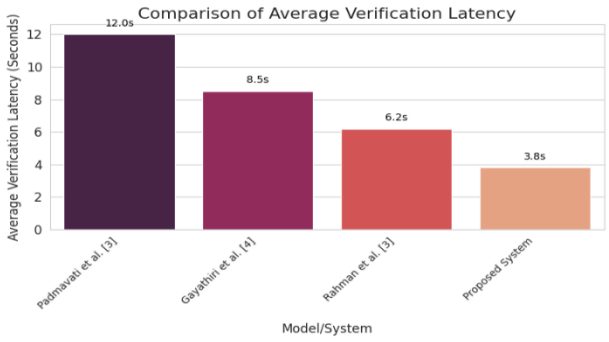


Figure 3. Comparison of Average Verification Latency

Further analysis of the system's classification ability using the Confusion Matrix as seen in Figure 4 which confirmed its exceptional ability to identify the positive class (invalid certificates). Out of the 200 samples tested, the system recorded 100 True Positives (TP) and 90 True Negatives (TN), resulting in only 10 False Positives (FP). Crucially, the system registered zero False Negatives (FN=0), meaning it successfully detected every single invalid certificate present in the test set. These zero false negative rates translate to perfect Recall for the positive class, which is vital for security applications where missing a threat is unacceptable. The overall discriminative power was quantified through standard classification curves. The ROC Curve as seen Figure 5 which yielded an AUC of 0.9804, indicating that the system's probability score for a randomly chosen positive instance is higher than that of a negative instance 98.04% of the time

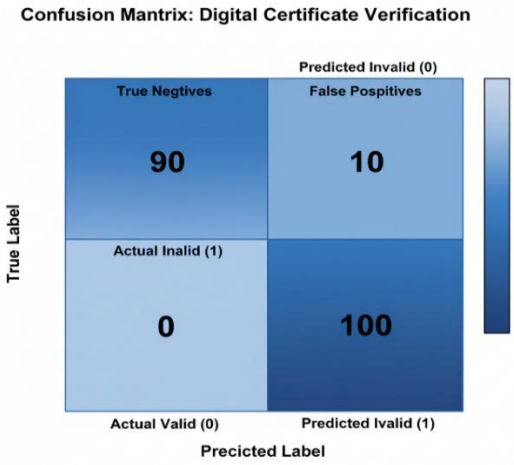


Figure 4. Confusion Matrix

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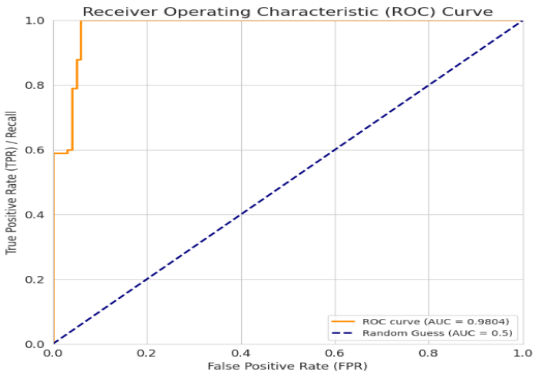


Figure 5. ROC Curve

Similarly, the Precision-Recall (PR) Curve as seen in Figure 6 which confirmed the model's performance on the positive class by achieving an Area Under the Curve (PR-AUC) of 0.9770.

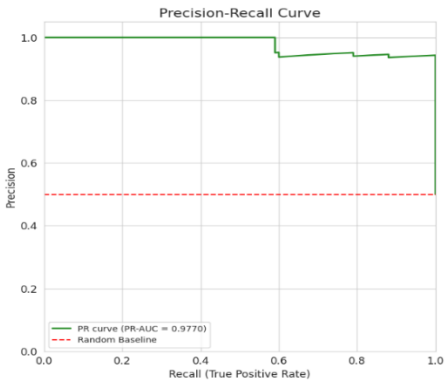


Figure 6. Precision-Recall Curve

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Both curve analyses, with AUC values extremely close to 1.0 and significantly above their respective random baselines, robustly demonstrate the high quality and reliability of the proposed system's classification capabilities.

IV. CONLCUSION

Our Proposed system delivers a Fraud Detection Accuracy of 97.2% with an Average Verification Latency of just 3.8 seconds, making it well-suited for real-time applications. The model ensures perfect recall (no false negatives) and demonstrates strong discriminative ability with ROC AUC of 0.9804 and PR-AUC of 0.9770. These results highlight its reliability, efficiency, and potential for secure digital certificate verification. Future work should focus on scalability, testing with larger and more diverse datasets, and exploring deployment on distributed frameworks like blockchain to maintain speed under heavy loads. Real-world applications could include e-commerce payment security or access control in critical infrastructure. Although only 10 false positives were observed, refining the model to reduce them without compromising recall remains an important direction. Additionally, strengthening resilience against adversarial attacks will be essential for long-term robustness.

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